

JS Advanced For Loops

1. Advanced For Loops in JavaScript:

- a. Advanced for loops in JavaScript are designed to simplify the process of iterating through collections like arrays, objects, strings, or other iterable entities.
- b. They provide cleaner syntax and functionality compared to traditional loops like **for** and **while** loop.
- c. **These loops divided into two types:**
 - i. for-in loop
 - ii. for-of loop

2. **for-in** Loop:

- a. The for-in loop is used to iterate over the **keys** (property names) of an object.
- b. It loops through all **enumerable properties** of an object, including inherited ones.
- c. **Meaning of Enumerable:** In JavaScript, **enumerable** means whether a property of an object can be **looped over** when using methods like **for...in** loop, **Object.keys()**.
- d. **Syntax:**



```
for (key in object) {  
    // Code to execute  
}
```

e. For Example →

```
const student = {  
  name: "John",  
  age: 22,  
  course: "React",  
};  
  
for (let key in student) {  
  console.log(key, ":", student[key]);  
}
```

3. **for-of** Loop:

- a. The for-of loop is used to iterate over iterable objects such as arrays, strings, maps, sets, or other iterable structures.
- b. It directly accesses the *values* of the iterable.
- c. **Syntax:**



```
for (value of iterable) {  
  // Code to execute  
}
```

d. For Example →

```
const colors = ["red", "green", "blue"];  
  
for (let color of colors) {  
  console.log(color);  
}
```

👉 So simply remember:

- 1. **for-in** loop → working with *objects*.
- 2. **for-of** loop → working with *arrays*, *strings* or *sets*, etc.